

**Language Doesn't Matter: Evaluating the Efficacy of Large Language Models in Analyzing
Sentiment in Non-English Textual Data**

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Abstract

This study examines whether general-purpose large language models (LLMs) can analyze sentiment in non-English text without task-specific fine-tuning, and whether prompt language affects their performance. We evaluate twelve open- and closed-weight LLMs on 200 French sentences drawn from a corpus of 2,683 French news articles on open-source software, comparing performance to dictionary-based methods across three conditions: French prompts on French text, English prompts on French text, and English prompts on English translations. Prompt language has no statistically meaningful effect on performance, and equivalence tests rule out any substantial English-prompting advantage, contradicting prior evidence of a cross-linguistic English-prompting advantage; translating the text into English, by contrast, lowers performance. Leading closed-weight models achieve strong performance ($r > 0.80$), larger open-weight models offer a viable, more accessible alternative, and the smallest model tested (1B parameters) fails to follow the task instructions. LLMs outperform a standard sentiment dictionary on correlation with human judgments and on coarse classification, though the dictionary offers greater transparency. All models perform substantially better on three-category (positive/neutral/negative) than on seven-category classification. These findings suggest social science researchers can use general-purpose LLMs for non-English sentiment analysis in whichever prompt language is most convenient, without sacrificing accuracy.

Keywords: Sentiment Analysis, Large Language models, Cross-lingual NLP, French Language, Open-Source models, Computational Social Science, Prompt Engineering

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Introduction

Most foundational large language models (LLMs) are trained predominantly on English data (Touvron et al., 2023), and a growing body of research reports that prompting in English improves performance on non-English tasks, even when the target text itself is non-English (Jiang et al., 2020; Lin et al., 2022; Muennighoff et al., 2023; Nie et al., 2024). If this “English prompt advantage” holds generally, it imposes a real cost on non-anglophone researchers, who would need to translate their prompts to get the best results from otherwise capable models.

This paper tests that assumption directly, using sentiment analysis of French-language text as a case study: **does the language of the prompt meaningfully affect an LLM’s performance, or does it not matter?** Sentiment analysis is well suited to this test because it demands cultural and linguistic nuance beyond simple translation, it is a common analytical task across the social sciences, it admits both continuous (intensity) and categorical (polarity) evaluation, and established dictionary-based methods offer a meaningful performance benchmark.

Answering this question well requires first establishing that general-purpose LLMs — used off the shelf, without task-specific fine-tuning — can do sentiment analysis on non-English text competently at all. Specialized models such as BERT variants achieve strong performance through fine-tuning (Anirban & Abhik, 2021; Michailidis, 2024; Rafeef A. Hameed et al., 2023), but require machine learning expertise, computational resources, and labeled data that many social science researchers lack (Chalkidis et al., 2020; Laurer et al., 2023; Strubell et al., 2019). General-purpose LLMs accessed through APIs offer a more accessible alternative, requiring primarily prompting skills (Ghosh et al., 2023; Perron et al., 2024) — but only if they perform adequately without fine-tuning in the first place.

We therefore ask two questions: (1) does prompt language (French vs. English) meaningfully affect sentiment analysis performance on French text, and (2) can general-purpose

LLMs — open and closed-weight¹ — perform this task well enough, without fine-tuning, to be a viable alternative to dictionary-based methods for social science research. We evaluate leading open and closed-source foundational LLMs on French news text and find that prompt language has no meaningful effect on performance, and that several models — including open-weight ones — perform competitively with or better than traditional dictionary methods.

Literature Review

Specialized, fine-tuned models achieve strong multilingual sentiment analysis performance — often above 85-95% accuracy across Indo-European, Semitic, and Asian languages (Ahmadian et al., 2024; Anirban & Abhik, 2021; ElJundi et al., 2019; Michailidis, 2024; Rafeef A. Hameed et al., 2023), including for lower-resource languages (Nazir et al., 2025; Salahudeen et al., 2023; Tela et al., 2024) and via cross-lingual transfer (Kumar & Albuquerque, 2021; Pan et al., 2023). This performance is costly: such models require large parameter counts (Francesco et al., 2021; Rafeef A. Hameed et al., 2023), fine-tuning expertise (Kotelnikova et al., 2023; Přibáň et al., 2024), and substantial labeled training data (Mikhail et al., 2024; Salahudeen et al., 2023) — barriers that put them out of reach for many social science researchers.

By contrast, general-purpose foundational LLMs used without fine-tuning remain understudied for multilingual sentiment analysis. Preliminary work with Mistral-7B, LLaMA2, and instruction-tuned models suggests promise in Italian, Russian, and cross-lingual settings (Buscemi & Proverbio, 2024; Plagwitz et al., 2024; Rusnachenko et al., 2024; Venerito & Iannone, 2024). Evidence on prompt language effects is mixed: some studies find language-specific prompting advantages (Hämäläinen et al., 2022; Vileikytė et al., 2024), while others report that English prompting broadly improves cross-linguistic performance (Jiang et al., 2020; Lin et al., 2022; Muennighoff et al., 2023; Nie et al., 2024). This study addresses this gap by evaluating leading open- and closed-weight foundational LLMs on French text.

¹ While “open source” traditionally implies access aligned with FOSS principles, its usage within the LLM context often refers more specifically to the public release of model weights, a condition more accurately described as “open weights.” This paper adopts this common LLM terminology while acknowledging that most models in our “open source” category are technically “open weights.”

Data & Methods

This study evaluates general-purpose large language models against dictionary-based baselines for sentiment analysis of French text, testing both prompt-language effects and overall LLM viability without task-specific fine-tuning.

Data Collection and Preparation

Our corpus consists of 2,683 French-language news articles from 13 media sources (9 from Québec, 1 from Ontario, 3 from France), an exhaustive collection of articles discussing open-source software from the Eureka database's "Major daily newspapers (FR)" category. Articles were selected using the search query below, spanning January 1, 1991 (the year of the first Linux kernel release) to January 1, 2025; the earliest article meeting this query was published June 23, 1995.

```
TEXT= ("logiciel libre" | "logiciels libres" | "open source" | "open-source" | "logiciel  
open source" | "logiciels open source" | "code source ouvert" | "software libre" |  
"free software" | "code source libre" | "Free Software Foundation" | "Richard  
Stallman")
```

Table 1 shows the distribution of articles by source media.

We built a Python/Selenium scraper to download articles in HTML format (rather than Eureka's default PDF batches, which are slow to retrieve and poorly suited to text analysis), then extracted publication metadata and full-text content with custom R functions.

Table 1
Summary of Collected News Articles Discussing Open Source Software by Source Media

source_media	article_count	earliest_date	latest_date
Libération	769	1995-06-23	2024-11-04
Monde, Le	666	2001-01-09	2024-12-26
Figaro, Le	402	1998-10-21	2024-12-04
Devoir, Le	318	1998-07-20	2024-10-26
Soleil, Le (Québec, QC)	179	1998-11-08	2023-11-18
Presse, La	152	1998-03-26	2017-10-14
Journal de Montréal, Le	65	2006-12-20	2023-04-24
Journal de Québec, Le	45	2007-03-24	2024-10-19
Tribune, La (Sherbrooke, QC)	43	2000-03-29	2022-12-03
Droit, Le (Ottawa, ON)	17	1998-11-10	2014-07-17
Nouvelliste, Le (Trois-Rivières, QC)	13	2000-01-13	2022-03-05
Quotidien, Le (Saguenay, QC)	10	2008-11-01	2019-04-10
Voix de l'Est, La (Granby, QC)	4	2010-09-17	2015-02-14

Figure 1
Temporal Distribution of News Articles Discussing Open Source Software

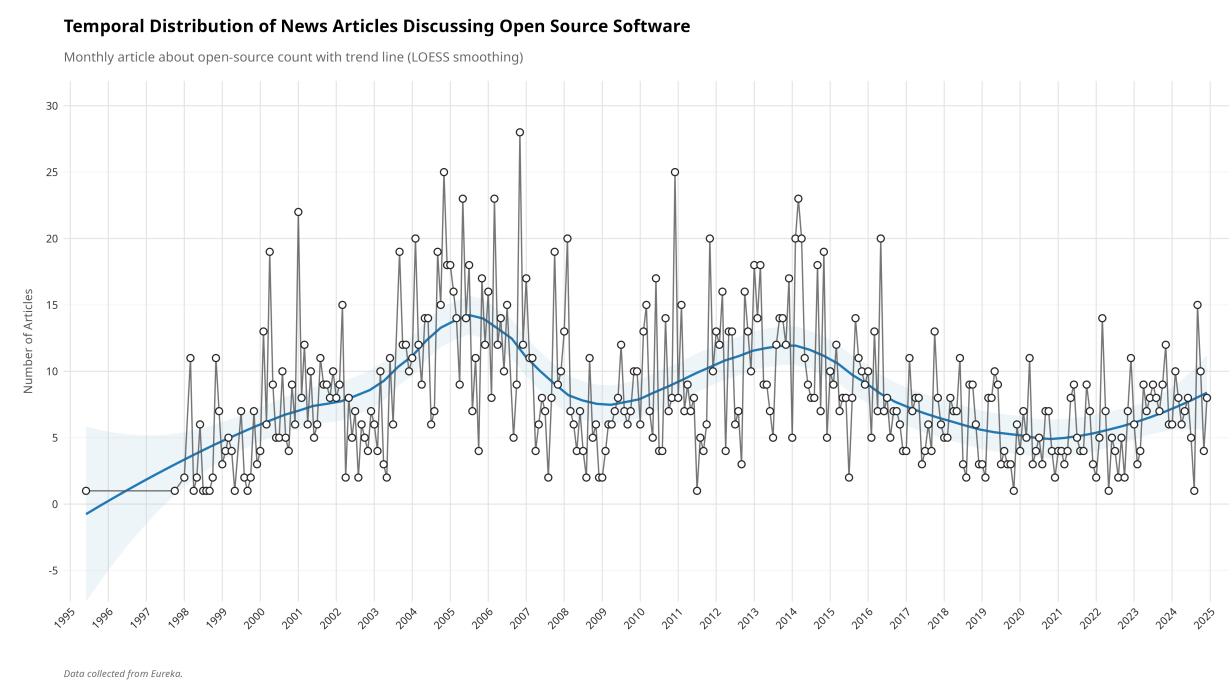


Figure 1 shows the temporal distribution of articles, with notable increases in coverage in the early 2000s corresponding to major open-source developments.

Translation

To enable the EN→EN condition described below, we translated the French corpus into English using Google Translate’s document translation service, processed in batches via an automated R workflow that preserved document IDs to keep French and English versions aligned in the final dataset.²

Manual Annotation

We created a validation benchmark of 200 sentences randomly sampled from the corpus, each translated into English via Google Translate and the `polyglotR` package. Three independent coders rated each sentence’s sentiment on a scale from -1 (very negative) to 1 (very positive) using a custom Shiny application with keyboard shortcuts and scale reminders; we averaged the three ratings per sentence to obtain the ground truth used to evaluate both dictionary-based and LLM methods.

Dictionary-Based Sentiment Analysis

We applied lexicon-based sentiment scoring to both the French original and English translation as a classical baseline. For French text, we used a customized version of the French Lexicoder Sentiment Dictionary (`fr1sd`) (Duval & Pétry, 2016) on lowercased, stopword-removed, sentence-segmented text, scoring each sentence as $(\text{positive terms} - \text{negative terms}) / \text{total words}$. For English translations, we used the standard Lexicoder Sentiment Dictionary (Young & Soroka, 2012) with additional handling for contractions, proper nouns, and negation (e.g., “not good” scored as negative), scoring each sentence as $((\text{positive} + \text{negated_negative}) - (\text{negative} + \text{negated_positive})) / \text{total words}$.

LLM-Based Sentiment Analysis

We evaluated twelve general-purpose foundational LLMs, selected for prominence and affordability — realistic choices for researchers who need to analyze large corpora rather than a single sample. Specialized fine-tuned models can outperform them on narrow tasks, at the cost in

² See `src/21_translate_to_english.R` in the project repository for the full translation pipeline.

expertise and infrastructure set out above. Tooling adds a further barrier that is rarely acknowledged: to our knowledge no R package exposes a fine-tuning workflow, so a researcher working in the language most common in quantitative social science must write and maintain training code themselves, whereas general-purpose models are reachable through an API with no comparable setup (Ghosh et al., 2023; Perron et al., 2024).

The evaluation spans nine open-weight models — Llama 3.2 1B, Llama 3.2 3B, Llama 3.1 8B, GPT-OSS 20B, Qwen3 32B, Llama 4 Scout, GPT-OSS 120B, Qwen3 235B-A22B, and DeepSeek V3.2, ranging from 1B to 671B total parameters — and three closed-weight models: Claude Haiku 4.5 (Anthropic), Gemini 3.5 Flash (Google), and GPT-5.6 Luna (OpenAI). This lineup updates an initial 2024-2025 evaluation of 11 comparable models using an identical experimental design; several models from that original evaluation (e.g., GPT-4o, Gemini 2.0 Flash) had already been deprecated by their providers within eighteen months, underscoring how quickly this landscape moves. The smallest model, Llama 3.2 1B, rarely followed the instruction to answer with a number: about one call in 23 returned a parseable score, and even after retries it produced a usable score for only 58 to 74 of the 200 sentences, depending on the condition. We retain it deliberately: while models of a few billion parameters can handle the task, a 1B-parameter model appears too small to follow the instructions reliably at all, which marks a practical lower bound on model size. Its metrics rest on a reduced and non-random subset of sentences and are not directly comparable to those of the other models.

For each model, we tested three conditions to isolate the effects of prompt language and text language: French prompts on French text (FR→FR), English prompts on French text (EN→FR), and English prompts on the English translation (EN→EN).

All prompts followed a standardized structure: a task definition, a five-point sentiment scale from -1.0 to 1.0 with anchor descriptions (allowing intermediate values), an instruction to consider cultural and linguistic nuance, and a constraint to respond with only a numerical value. French prompts were direct translations of the English prompts. The English-prompt/French-text

version is shown below.³

```
en_prompt_fr_text <- function(text) {
  paste0("Please analyze the sentiment of the following French text and provide a single
  numerical rating according to this scale:
  Sentiment Scale:
  -1.0: Strong negative sentiment - highly critical, hostile, or pessimistic content
  -0.5: Moderate negative sentiment - somewhat negative, disapproving, or concerned
  content
  0.0: Neutral sentiment - factual, balanced, or neither positive nor negative content
  0.5: Moderate positive sentiment - somewhat positive, approving, or optimistic content
  1.0: Strong positive sentiment - highly supportive, enthusiastic, or optimistic content
  You may also use values between these points for intermediate sentiment levels
  (e.g., -0.7, 0.3).
  Important instructions:
  1. First, carefully read and understand the text, considering cultural and linguistic
  nuances in French.
  2. Analyze the emotional tone, word choice, and overall message.
  3. Respond ONLY with a single numerical value between -1.0 and 1.0 that best
  represents the sentiment.
  4. Do not include ANY explanations, analysis, or additional text in your response.
  Here is the text to analyze: ", text)
}
```

We queried each model via the `ellmer` R package, using the vendors' own APIs for the three closed-weight models and OpenRouter for the nine open-weight models, with responses capped at 400 tokens. Because OpenRouter can serve the same model through several providers at different numerical precisions, each open-weight model was pinned to a single named provider endpoint, at the highest precision offered, with fallback to other providers disabled.⁴ Each sentence was scored three times per model-condition combination (21,600 scores in total), and we took the mean of the three runs as the final score. Despite explicit instructions to respond with a bare number, models frequently returned formatted text (e.g., "Sentiment: 0.5"); we parsed these

³ The French-prompt/French-text version is identical except for language; see `src/93_prompts.R` in the project repository for the exact prompts, and `src/40_prompt.R` for the full evaluation pipeline, including model initialization, retry logic, rate limiting, and checkpointing.

⁴ Endpoint pins, and the reasoning settings used to suppress chain-of-thought in models that reason by default, are listed in `src/94_models_map.R`.

with a pattern-matching function validated against the expected -1.0 to 1.0 range.

Validation and Performance Evaluation

We assessed performance with three complementary metrics: Pearson correlation between each model's continuous predictions and the manually annotated ground truth; weighted F1 score on a 7-category classification ("very_negative" through "very_positive"); and weighted F1 score on a simplified 3-category classification (negative/neutral/positive) that groups the 7 categories into broad polarity bands. We use weighted (rather than macro) F1 because the manually coded categories are unevenly distributed — "somewhat_positive" alone accounts for 44% of annotated sentences, while "very_negative" and "very_positive" each account for 1% or less — so a weighted average better reflects overall classification quality given this class imbalance. Comparing the two classification schemes lets us separately assess fine-grained sentiment discrimination (7-category) and broad polarity classification (3-category). We additionally report mean absolute error (MAE) between continuous predictions and the ground truth. All metrics were calculated on complete cases only.

Results

We evaluated models using the three metrics described above, complemented by mean absolute error (MAE). The condition analysis and the averages reported below cover the eleven models that scored all 200 sentences (33 model-condition combinations) and exclude the dictionary baselines, which have no prompt. Llama 3.2 1B is reported separately: it scored a different, partial subset of sentences in each condition, so its differences across conditions would compare unlike samples.

Prompt Language Has No Meaningful Effect; Translating the Text Does

Holding the text constant, prompt language made no detectable difference. French prompts on French text (FR→FR) edged out English prompts on the same text (EN→FR) on every metric — mean correlation 0.700 vs. 0.673, MAE 0.209 vs. 0.217, 3-category F1 0.669 vs. 0.646, 7-category F1 0.364 vs. 0.347 — but none of these paired differences was statistically significant (all $p > 0.12$). Because a non-significant test cannot on its own establish the absence of

an effect, we also ran two one-sided tests (TOST) against an equivalence bound of ± 0.05 on each metric's scale. FR→FR and EN→FR were statistically equivalent on MAE ($p = 0.002$), 7-category F1 ($p = 0.034$), and 3-category F1 ($p = 0.034$). Correlation was the one metric where equivalence could not be established: the 90% confidence interval of the difference, $[-0.032, 0.085]$, leaves room for a French-prompt advantage of up to 0.085, but not for an English-prompt advantage larger than 0.032.

Translating the text into English, by contrast, had a measurable cost. Relative to FR→FR, scoring the English translation (EN→EN) lowered mean correlation by 0.053 (0.700 vs. 0.647, $p = 0.007$) and 3-category F1 by 0.063 (0.669 vs. 0.606, $p < 0.001$); the 3-category loss also held against EN→FR (0.041, $p = 0.005$). Differences in MAE and 7-category F1 across all three conditions remained within the ± 0.05 equivalence bound. The translation penalty is therefore concentrated in how well models rank sentences by sentiment and identify their polarity, consistent with information lost in machine translation.

LLMs Are a Viable Alternative to Dictionary Methods

General-purpose LLMs, used without fine-tuning, correlated with human judgments well above the dictionary baselines. The best configurations exceeded $r = 0.80$: GPT-5.6 Luna led on correlation ($r = 0.810$ in FR→FR), Claude Haiku 4.5 on both 3-category F1 (0.778) and 7-category F1 (0.537, both FR→FR), and Gemini 3.5 Flash on MAE (0.156 in EN→FR). Nine of the 33 model-condition combinations correlated with the ground truth more strongly than individual human coders agreed with the mean of the other two (mean leave-one-out $r = 0.748$, range 0.724–0.779).⁵

⁵ This comparison favors the models slightly, since the ground truth they are compared against averages three coders rather than two.

Figure 2
Sentiment Analysis Models Correlation with Ground Truth

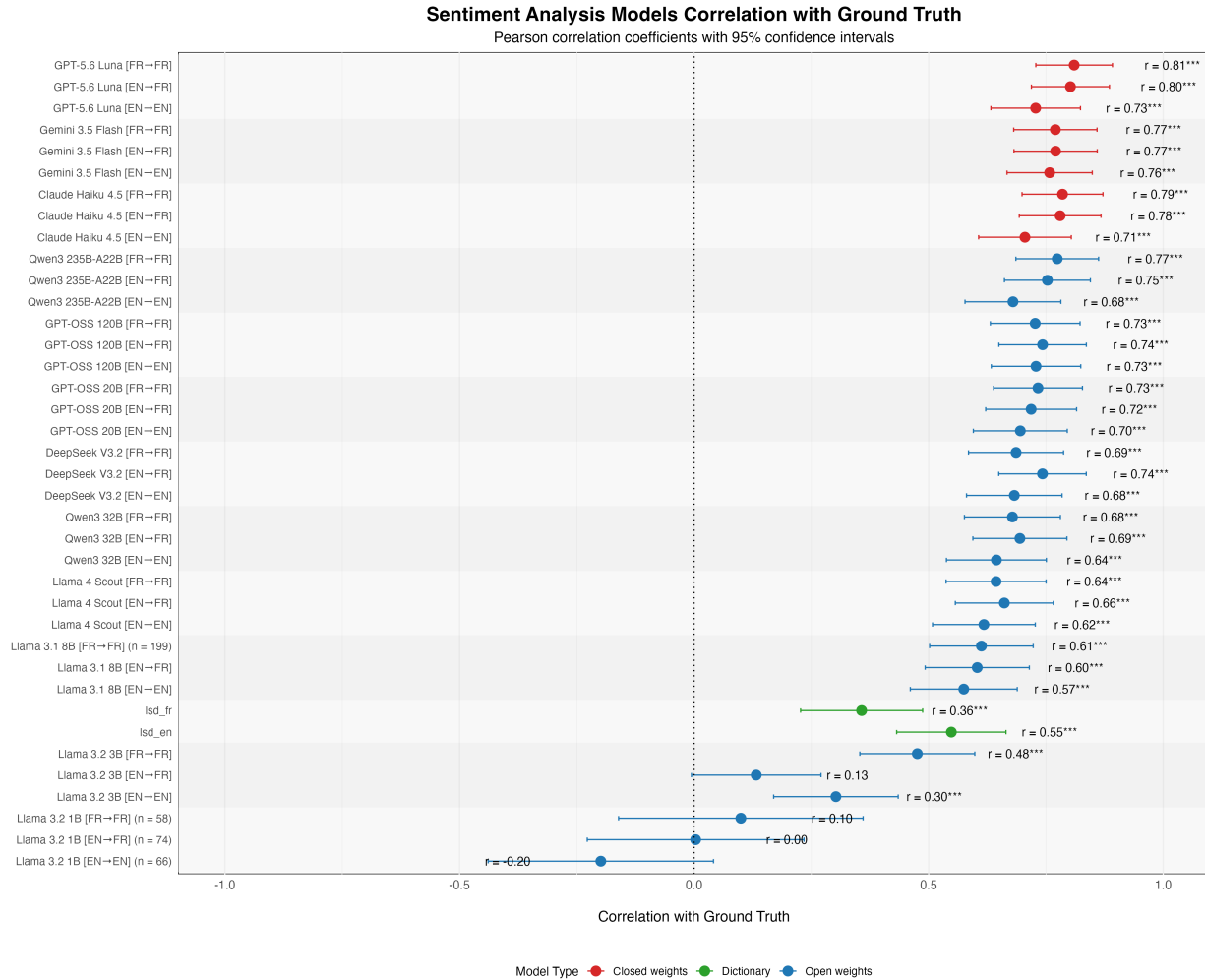


Figure 1. Cross-lingual sentiment analysis performance across model architectures. Models grouped by type (color) and sorted by correlation strength. Prompting mechanisms within clusters ordered as FR→FR (French prompt on French text), EN→FR (English prompt on French text), and EN→EN (English prompt on translated text). Error bars represent 95% confidence intervals; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Closed-weight models outperformed open-weight ones on average (mean $r = 0.768$ vs. 0.638; 3-category F1 0.689 vs. 0.622), but the gap closed at the top of the open-weight range: Qwen3 235B-A22B reached $r = 0.774$ in FR→FR, above Gemini 3.5 Flash in the same condition, and GPT-OSS 120B and GPT-OSS 20B averaged $r = 0.733$ and 0.716 across conditions. Performance degraded steadily below that. Llama 3.1 8B remained usable ($r = 0.575$ –0.612), Llama 3.2 3B was erratic ($r = 0.476$ in FR→FR but 0.132 in EN→FR), and Llama 3.2 1B failed outright: it returned a usable score for only 58 to 74 sentences per condition, and on those its correlation with the ground truth ranged from -0.199 to 0.100.

The English Lexicoder Sentiment Dictionary reached a correlation of 0.548 and the French version 0.357, against a mean of 0.673 for the eleven LLMs; on 3-category F1 the dictionaries (0.588 and 0.557) also trailed the LLM mean (0.640), though the English dictionary outperformed Llama 3.2 3B and Qwen3 32B. On MAE, however, both dictionaries (0.203 and 0.210) did slightly better than the LLM average (0.213). This reflects the dictionaries' compressed output rather than accuracy: their scores cluster near zero ($SD = 0.065$ for the English dictionary, vs. 0.286 for the ground truth and 0.38–0.47 for the LLMs), which keeps absolute errors small on a sample dominated by mildly positive sentences while discriminating poorly between them.

Classification Granularity

All models performed substantially better on 3-category than 7-category classification (best F1 of 0.778 vs. 0.537; mean 0.640 vs. 0.349; see Figure 3, below). Models most reliably identified positive and negative sentiment (7-category F1 = 0.482 and 0.441, rising to 0.690 and 0.627 in the 3-category scheme) and struggled most with the extreme categories (very_negative F1 = 0.256, very_positive F1 = 0.124), which are nearly absent from the sample (two sentences and one sentence respectively). Neutral remained the hardest 3-category label (F1 = 0.258). Researchers needing fine-grained sentiment distinctions may find general-purpose LLMs insufficient without further adaptation; the broader positive/neutral/negative distinction is handled far more reliably.

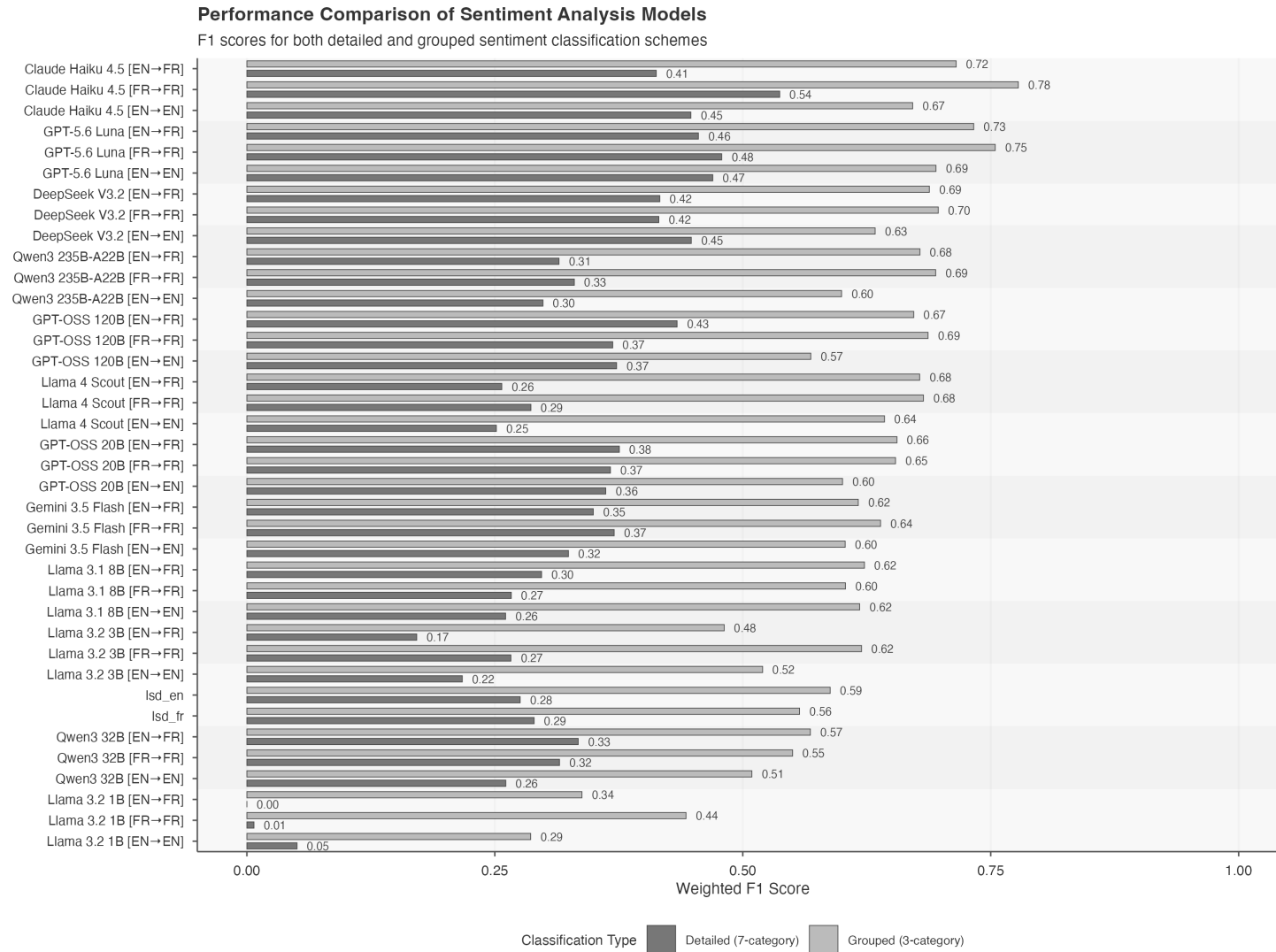


Figure 2. F1 scores for sentiment analysis across model architectures. Models grouped by F1 score performance and sorted by 3-category classification. Prompting mechanisms: EN→FR (English prompt on French text), FR→FR (French prompt on French text), and EN→EN (English prompt on translated text). Detailed classification includes 7 categories, grouped classification uses 3.

Figure 3
F1 Scores Comparison Across Models and Categories

Discussion

Our central finding is that prompt language does not meaningfully affect LLM sentiment analysis performance on French text — a more nuanced picture than the English-prompting advantage reported elsewhere ([Jiang et al., 2020](#); [Lin et al., 2022](#); [Muennighoff et al., 2023](#)). If anything, the point estimates favored French prompts, and equivalence tests ruled out any English-prompt advantage larger than a few hundredths on every metric. What did cost performance was translating the text itself: scoring English translations lowered correlation and 3-category F1 relative to working with the French original. This challenges the assumption that researchers must work in English to get good results: francophone researchers can use their native language with no analytical penalty, teams can choose prompt language by convenience rather than performance, and multi-national collaborations can let each team prompt in its preferred language while keeping results comparable — without needing to standardize on English. The practical warning runs the other way: translating a corpus into English before analysis, a common step in computational work on non-English text, is not neutral. Theoretically, this suggests the “English prompt advantage” documented in prior work may be task-dependent, language-pair specific, or diminishing as models improve.

This finding is only practically useful if LLMs are themselves viable for the task, and our results support that they are. Leading closed-weight models achieved strong correlations (GPT-5.6 Luna, $r = 0.810$) and 3-category F1 scores (Claude Haiku 4.5, 0.778) without any task-specific fine-tuning, several configurations agreed with the ground truth more closely than our individual human coders did, and the largest open-weight models came close to them (Qwen3 235B-A22B, $r = 0.774$). Model size sets a floor, however. Models from roughly 8B parameters upward produced usable scores, a 3B model was unreliable across conditions, and a 1B model could not follow the instruction to return a number at all. Together, these results mean social scientists can confidently deploy leading closed-weight LLMs, and larger open-weight ones, for French sentiment analysis in whichever prompt language suits their workflow, without fine-tuning expertise or heavy computational resources ([Přibáň et al., 2024](#)) — provided they avoid the smallest models.

Two secondary results qualify this picture. First, LLMs clearly outperformed the Lexicoder Sentiment Dictionary on correlation (mean LLM $r = 0.673$ vs. 0.548 for the English dictionary) and, more modestly, on 3-category F1 (0.640 vs. 0.588), but the dictionaries' near-zero output gave them a slightly lower MAE — a reminder that MAE alone can reward a method that barely discriminates. A transparent, fully interpretable dictionary remains a defensible baseline for basic polarity detection, but not for measuring sentiment intensity. Second, all models performed markedly worse on fine-grained 7-category classification than on 3-category polarity (mean F1 of 0.349 vs. 0.640), struggling most with extreme sentiment categories; researchers needing granular sentiment distinctions should not expect general-purpose LLMs to deliver this out of the box, though Claude Haiku 4.5's stronger 7-category performance (0.537) suggests some models handle nuance better than others.

Limitations

Several limitations qualify these findings. The 200-sentence validation sample, though randomly drawn, is a small fraction of the corpus and may not capture its full linguistic range; the corpus itself is French-language news on a single technical/political domain (open-source software), so performance hierarchies and cross-lingual prompting effects may not extrapolate to other languages, especially those less represented in LLM training data, or to other genres such as social media or political debate text with different linguistic styles.

Methodologically, English translations relied on Google Translate without a formal fidelity check, so translation artifacts could have affected the EN→EN condition specifically. Model selection prioritized affordability and prominence over including every top-performing model globally, reflecting the perspective of researchers who need to analyze corpora of millions of sentences rather than a few hundred. A single standardized prompt was used across all models for comparability, which may understate peak performance achievable through model-specific prompt engineering; the need for a response-parsing function also indicates models did not always follow the requested output format exactly.

Finally, no cost or runtime data were collected, though this information matters for

researchers assessing feasibility. More fundamentally, this evaluation is a snapshot: given how quickly LLM capabilities and provider offerings change — several models evaluated in an earlier version of this study were deprecated within eighteen months — the specific rankings reported here should be expected to age quickly, even as the higher-level finding regarding prompt language is likely to hold across model generations.

Conclusion

Prompt language does not constrain how social scientists work with non-English text. The practical consequence runs further than francophone teams being free to prompt in French: prompt language ceases to be a design decision that needs defending at all, and the translation step that has often preceded computational analysis of non-English corpora loses its rationale.

That opens work this study could not do. Comparative projects spanning several languages can let each team prompt in its own and pool the results, without a translation pass that would itself require validation. Whether this holds beyond French — particularly for languages less represented in training data, where the English-prompting advantage was first reported — is the natural next test, and this design ports to it directly: the same three conditions, the same manually annotated benchmark, a different target language.

Method choice remains a judgment about research goals rather than a ranking. Dictionary methods retain an advantage these results do not overturn — every score they produce can be traced to the words that generated it — and that transparency is worth more than accuracy in some research designs. What the evidence here settles is narrower and, for non-anglophone researchers, more useful: the language of the prompt is not among the things that need to be traded off.

References

- Ahmadian, H., Abidin, T. F., Riza, H., & Muchtar, K. (2024). Hybrid Models for Emotion Classification and Sentiment Analysis in Indonesian Language. *Applied Computational Intelligence and Soft Computing*, 2024(1). <https://doi.org/10.1155/2024/2826773>
- Anirban, B., & Abhik, J. (2021). Sentiment Analysis For Bengali Using Transformer Based Models. *ICON*.

- Buscemi, A., & Proverbio, D. (2024). ChatGPT vs Gemini vs LLaMA on Multilingual Sentiment Analysis. *arXiv.org*. <https://doi.org/10.48550/ARXIV.2402.01715>
- Chalkidis, I., Fergadiotis, M., Malakasiotis, P., Aletras, N., & Androutsopoulos, I. (2020). LEGAL-BERT: The Muppets straight out of Law School. *Findings of the Association for Computational Linguistics: EMNLP 2020*.
<https://doi.org/10.18653/v1/2020.findings-emnlp.261>
- Duval, D., & Pétry, F. (2016). L'analyse automatisée du ton médiatique : Construction et utilisation de la version française du Lexicoder Sentiment Dictionary. *Canadian Journal of Political Science/Revue Canadienne de Science Politique*, 49(2), 197–220.
<https://doi.org/10.1017/S000842391600055X>
- ElJundi, O., Antoun, W., El Droubi, N., Hajj, H., El-Hajj, W., & Shaban, K. (2019). hULMonA: The Universal Language Model in Arabic. *Proceedings of the Fourth Arabic Natural Language Processing Workshop*, 68–77. <https://doi.org/10.18653/v1/w19-4608>
- Francesco, B., Luis, E. A., & José, C.-C. (2021). XLM-T: Multilingual Language Models in Twitter for Sentiment Analysis and Beyond. *International Conference on Language Resources and Evaluation*.
- Ghosh, A., Biswas, D., Sarkar, R., Pillai, A., & Chopra, J. (2023). Navigating GPT-4 and BERT: A Dual Perspective on Financial and Political Sentiment Analysis. *International Journal for Research in Applied Science and Engineering Technology*, 11(12), 1577–1583.
<https://doi.org/10.22214/ijraset.2023.57687>
- Hämäläinen, M., Alnajjar, K., & Poibeau, T. (2022). Video Games as a Corpus: Sentiment Analysis using Fallout New Vegas Dialog. *Proceedings of the 17th International Conference on the Foundations of Digital Games*, 1–4.
<https://doi.org/10.1145/3555858.3555930>
- Jiang, Z., Xu, F. F., Araki, J., & Neubig, G. (2020). How Can We Know What Language Models Know? *Transactions of the Association for Computational Linguistics*, 8, 423–438.
https://doi.org/10.1162/tac1_a_00324

- Kotelnikova, A. V., Vychezhnanin, S. V., & Kotelnikov, E. V. (2023). Cross-Domain Sentiment Analysis Based on Small in-Domain Fine-Tuning. *IEEE Access : Practical Innovations, Open Solutions*, 11, 41061–41074. <https://doi.org/10.1109/access.2023.3269720>
- Kumar, A., & Albuquerque, V. H. C. (2021). Sentiment Analysis Using XLM-R Transformer and Zero-shot Transfer Learning on Resource-poor Indian Language. *ACM Transactions on Asian and Low-Resource Language Information Processing*, 20(5), 1–13. <https://doi.org/10.1145/3461764>
- Laurer, M., van Atteveldt, W., Casas, A., & Welbers, K. (2023). Less Annotating, More Classifying: Addressing the Data Scarcity Issue of Supervised Machine Learning with Deep Transfer Learning and BERT-NLI. *Political Analysis*, 32(1), 84–100. <https://doi.org/10.1017/pan.2023.20>
- Lin, X. V., Mihaylov, T., Artetxe, M., Wang, T., Chen, S., Simig, D., Ott, M., Goyal, N., Bhosale, S., Du, J., Pasunuru, R., Shleifer, S., Koura, P. S., Chaudhary, V., O'Horo, B., Wang, J., Zettlemoyer, L., Kozareva, Z., Diab, M., ... Li, X. (2022). Few-shot Learning with Multilingual Generative Language Models. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. <https://doi.org/10.18653/v1/2022.emnlp-main.616>
- Michailidis, P. D. (2024). A Comparative Study of Sentiment Classification Models for Greek Reviews. *Big Data and Cognitive Computing*, 8(9), 107. <https://doi.org/10.3390/bdcc8090107>
- Mikhail, K., O., K., Liliana, C. H., Grigori, S., & A., G. (2024). Multilingual Approaches to Sentiment Analysis of Texts in Linguistically Diverse Languages: A Case Study of Finnish, Hungarian, and Bulgarian. *IWCLUL*.
- Muennighoff, N., Wang, T., Sutawika, L., Roberts, A., Biderman, S., Le Scao, T., Bari, M. S., Shen, S., Yong, Z. X., Schoelkopf, H., Tang, X., Radev, D., Aji, A. F., Almubarak, K., Albanie, S., Alyafeai, Z., Webson, A., Raff, E., & Raffel, C. (2023). Crosslingual Generalization through Multitask Finetuning. *Proceedings of the 61st Annual Meeting of*

the Association for Computational Linguistics (Volume 1: Long Papers).

<https://doi.org/10.18653/v1/2023.acl-long.891>

Nazir, M. K., Faisal, C. N., Habib, M. A., & Ahmad, H. (2025). Leveraging Multilingual Transformer for Multiclass Sentiment Analysis in Code-Mixed Data of Low-Resource Languages. *IEEE Access : Practical Innovations, Open Solutions*, 13, 7538–7554.

<https://doi.org/10.1109/access.2025.3527710>

Nie, E., Yuan, S., Ma, B., Schmid, H., Färber, M., Kreuter, F., & Schütze, H. (2024). Decomposed Prompting: Unveiling Multilingual Linguistic Structure Knowledge in English-Centric Large Language Models. *arXiv.org*. <https://doi.org/10.48550/ARXIV.2402.18397>

Pan, R., García-Díaz, J. A., Garcia-Sanchez, F., & Valencia-García, R. (2023). Evaluation of transformer models for financial targeted sentiment analysis in Spanish. *PeerJ Computer Science*, 9, e1377. <https://doi.org/10.7717/peerj-cs.1377>

Perron, B. E., Luan, H., Qi, Z., Victor, B. G., & Goyal, K. (2024). Demystifying Application Programming Interfaces (APIs): Unlocking the Power of Large Language Models and Other Web-based AI Services in Social Work Research. *arXiv.org*.

<https://doi.org/10.48550/ARXIV.2410.20211>

Plagwitz, L., Neuhaus, P., Yildirim, K., Losch, N., Varghese, J., & Büscher, A. (2024). Zero-Shot LLMs for Named Entity Recognition: Targeting Cardiac Function Indicators in German Clinical Texts. In *Studies in Health Technology and Informatics*. IOS Press.

<https://doi.org/10.3233/shti240861>

Přibáň, P., Šmíd, J., Steinberger, J., & Mištera, A. (2024). A comparative study of cross-lingual sentiment analysis. *Expert Systems with Applications*, 247, 123247.

<https://doi.org/10.1016/j.eswa.2024.123247>

Rafeef A. Hameed, Wael J. Abed, & Ahmed T. Sadiq. (2023). Evaluation of Hotel Performance with Sentiment Analysis by Deep Learning Techniques. *International Journal of Interactive Mobile Technologies (iJIM)*, 17(09), 70–87.

<https://doi.org/10.3991/ijim.v17i09.38755>

- Rusnachenko, N., Golubev, A., & Loukachevitch, N. (2024). Large Language Models in Targeted Sentiment Analysis. *arXiv.org*. <https://doi.org/10.48550/ARXIV.2404.12342>
- Salahudeen, S. A., Lawan, F. I., Wali, A. M., Imam, A. A., Shuaibu, A. R., Yusuf, A., Rabi, N. B., Bello, M., Adamu, S. U., Aliyu, S. M., Gadanya, M. S., Muaz, S. A., Ahmad, M. S., Abdullahi, A., & Jamoh, A. Y. (2023). HausaNLP at SemEval-2023 Task 12: Leveraging African Low Resource TweetData for Sentiment Analysis. *International Workshop on Semantic Evaluation*. <https://doi.org/10.48550/ARXIV.2304.13634>
- Strubell, E., Ganesh, A., & McCallum, A. (2019). Energy and Policy Considerations for Deep Learning in NLP. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. <https://doi.org/10.18653/v1/p19-1355>
- Tela, A., Woubie, A., & Hautamäki, V. (2024). Transferring monolingual model to low-resource language: The case of Tigrinya. *Applied Computing and Intelligence*, 4(2), 184–194. <https://doi.org/10.3934/aci.2024011>
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., Rodriguez, A., Joulin, A., Grave, E., & Lample, G. (2023). *LLaMA: Open and Efficient Foundation Language Models* (arXiv:2302.13971). arXiv. <https://arxiv.org/abs/2302.13971>
- Venerito, V., & Iannone, F. (2024). Large language model-driven sentiment analysis for facilitating fibromyalgia diagnosis. *RMD Open*, 10(2), e004367. <https://doi.org/10.1136/rmdopen-2024-004367>
- Vileikytė, B., Lukoševičius, M., & Stankevičius, L. (2024). Sentiment Analysis of Lithuanian Online Reviews Using Large Language Models. *International Conference on Information Technology*. <https://doi.org/10.48550/ARXIV.2407.19914>
- Young, L., & Soroka, S. (2012). Affective News: The Automated Coding of Sentiment in Political Texts. *Political Communication*, 29(2), 205–231. <https://doi.org/10.1080/10584609.2012.671234>

Appendix A

Appendix B

Figure B1
Sentiment Analysis Models Mean Absolute Error

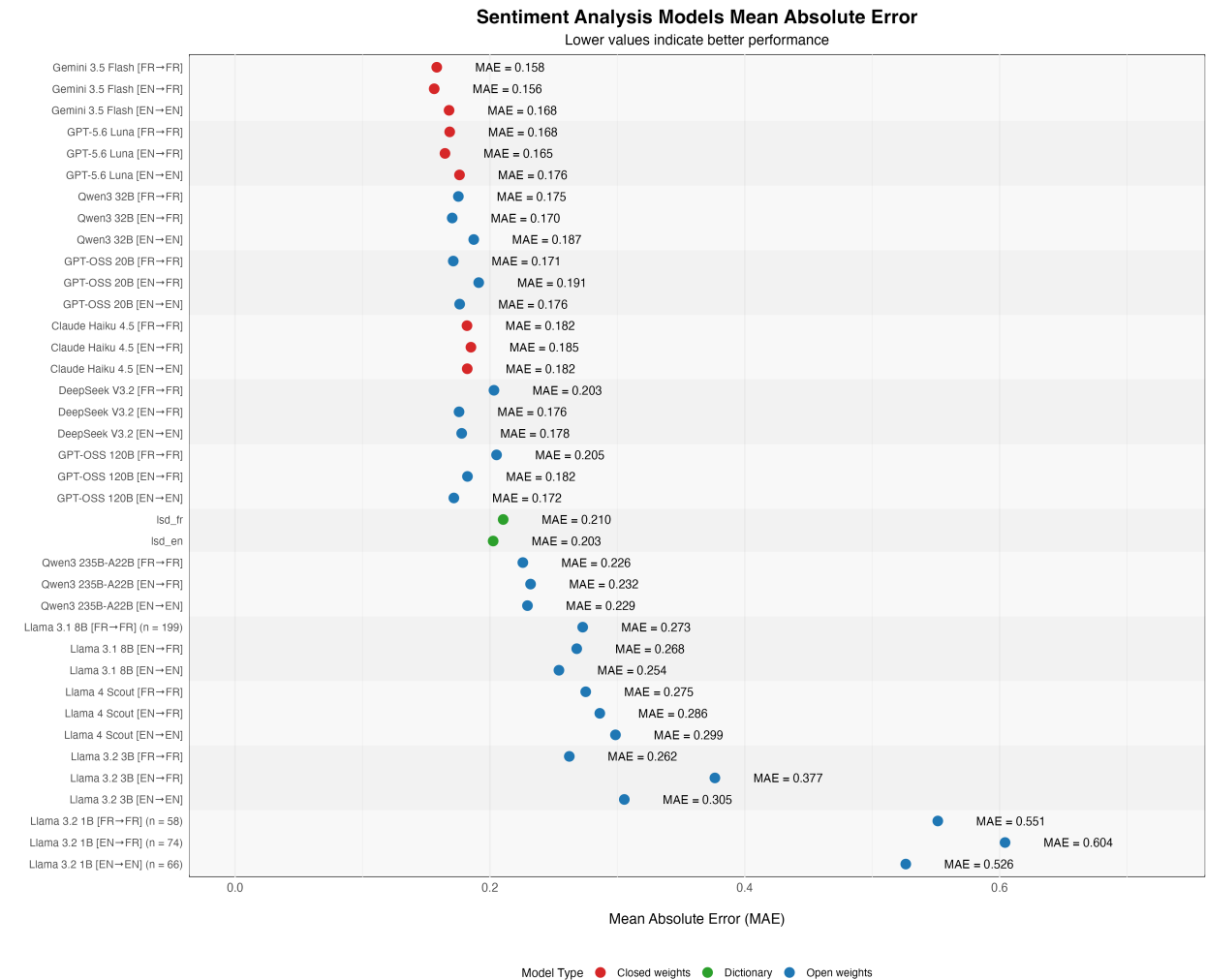


Figure 2. Cross-lingual sentiment analysis performance by Mean Absolute Error. Models grouped by type (color) and sorted by MAE (lower is better). Prompting mechanisms within clusters ordered as FR→FR (French prompt on French text), EN→FR (English prompt on French text), and EN→EN (English prompt on translated text).

Figure 4 shows the Mean Absolute Error (MAE) results across models and conditions, providing complementary insights into prediction accuracy beyond correlation analysis.

Figure B2
Sentiment Analysis Comparison Over Time

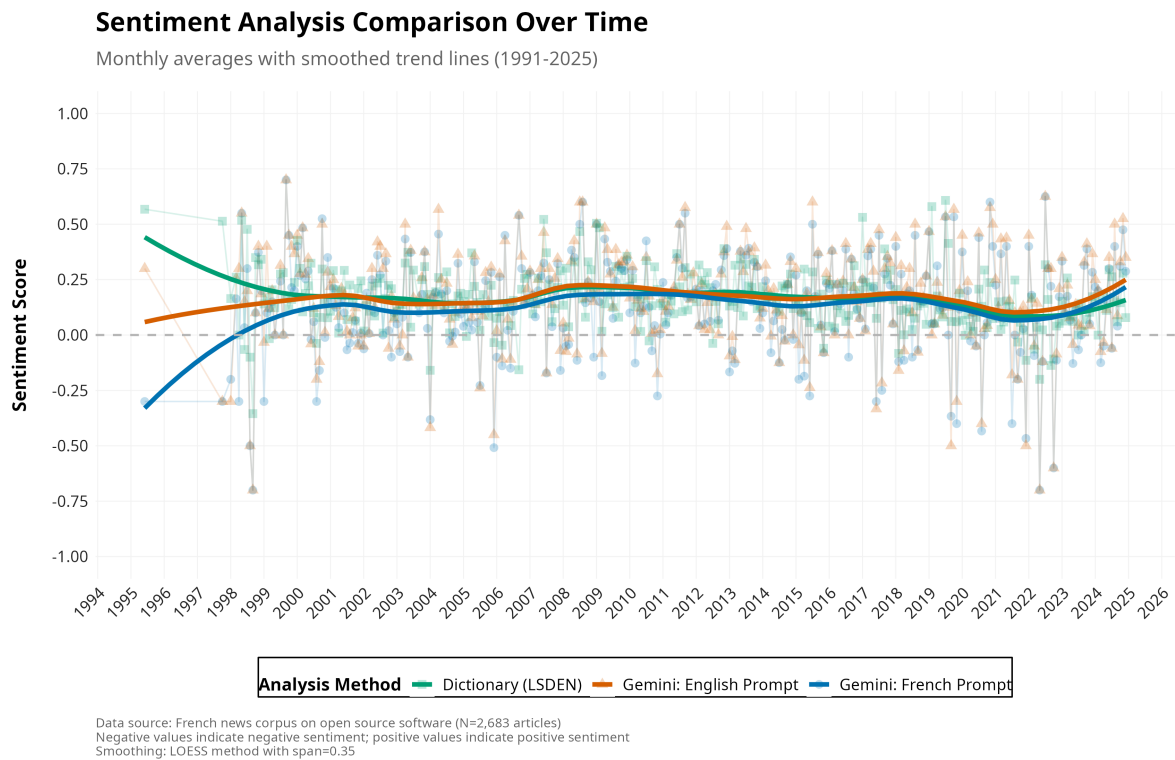


Figure 5 compares corpus-wide temporal sentiment trends captured by the best-performing LLM and by the Lexicoder Sentiment Dictionary, illustrating that the two methods track closely despite the LLM's higher correlation with sentence-level manual annotations.

Table B1*F1 Scores for 3-Category Classification*

Model	negative	neutral	positive	weighted_f1
claudehaiku45_fr_fr	0.762	0.375	0.832	0.778
gpt56luna_fr_fr	0.748	0.385	0.8	0.755
gpt56luna_en_fr	0.73	0.364	0.776	0.733
claudehaiku45_en_fr	0.69	0.273	0.778	0.715
deepseekv32_fr_fr	0.672	0.216	0.765	0.697
gpt56luna_en_en	0.688	0.367	0.736	0.695
qwen3235b_fr_fr	0.752	0.243	0.716	0.695
deepseekv32_en_fr	0.696	0.226	0.737	0.688
gptoss120b_fr_fr	0.721	0.314	0.712	0.687
llama4scout_fr_fr	0.643	0.291	0.747	0.682
qwen3235b_en_fr	0.719	0.254	0.706	0.679
llama4scout_en_fr	0.667	0.258	0.732	0.678
gptoss120b_en_fr	0.717	0.324	0.689	0.672
claudehaiku45_en_en	0.678	0.256	0.715	0.671
gptoss20b_en_fr	0.7	0.268	0.677	0.655
gptoss20b_fr_fr	0.691	0.258	0.68	0.654
llama4scout_en_en	0.647	0.246	0.686	0.643
gemini35_fr_fr	0.655	0.264	0.673	0.639
deepseekv32_en_en	0.672	0.258	0.657	0.633
llama318b_en_fr	0.652	0.281	0.646	0.623
llama323b_fr_fr	0.465	0.195	0.747	0.62
llama318b_en_en	0.627	0.265	0.653	0.618
gemini35_en_fr	0.642	0.242	0.646	0.616
llama318b_fr_fr	0.615	0.174	0.643	0.604
gemini35_en_en	0.667	0.233	0.613	0.603
gptoss20b_en_en	0.614	0.234	0.635	0.601
qwen3235b_en_en	0.698	0.227	0.591	0.6
lsd_en	0.574	0.231	0.636	0.588
gptoss120b_en_en	0.678	0.218	0.552	0.569
qwen332b_en_fr	0.554	0.165	0.621	0.568
lsd_fr	0.474	0.289	0.631	0.557
qwen332b_fr_fr	0.515	0.177	0.611	0.55
llama323b_en_en	0.167	0.261	0.73	0.52
qwen332b_en_en	0.559	0.19	0.52	0.509
llama323b_en_fr	0	0.222	0.757	0.481
llama321b_fr_fr	0	0.138	0.648	0.443
llama321b_en_fr	0	0	0.568	0.338
llama321b_en_en	0	0.15	0.457	0.286

Table B2*F1 Scores for 7-Category Classification*

Model	very_neg	negative	somewhat_neg	neutral	somewhat_pos	positive	very_pos	weighted
claudehaiku45-fr-fr	0.4	0.68	0.455	0.375	0.54	0.612	0	0.537
gpt56luna-fr-fr	0.286	0.5	0.393	0.385	0.538	0.462	0.133	0.479
gpt56luna-en-en	0.364	0.5	0.338	0.367	0.51	0.542	0.25	0.47
gpt56luna-en-fr	0.333	0.5	0.258	0.364	0.564	0.407	0.125	0.455
deepseekv32-en-en	0.333	0.533	0.419	0.258	0.415	0.588	0.667	0.448
claudehaiku45-en-en	0.286	0.642	0.4	0.256	0.409	0.568	0	0.448
gptoss120b-en-fr	0.333	0.566	0.4	0.324	0.416	0.485	0.133	0.434
deepseekv32-en-fr	0.444	0.667	0.429	0.226	0.349	0.493	0	0.416
deepseekv32-fr-fr	0.333	0.368	0.469	0.216	0.4	0.527	0.167	0.415
claudehaiku45-en-fr	0.5	0.68	0.364	0.273	0.347	0.511	0.2	0.413
gptoss20b-en-fr	0.235	0.431	0.269	0.268	0.37	0.535	0	0.376
gptoss120b-en-en	0.444	0.692	0.281	0.218	0.288	0.536	0.286	0.373
gemini35-fr-fr	0.222	0.583	0.25	0.264	0.31	0.571	0.2	0.37
gptoss120b-fr-fr	0.308	0.545	0.333	0.314	0.319	0.447	0.154	0.369
gptoss20b-fr-fr	0.364	0.6	0.286	0.258	0.325	0.448	0.286	0.367
gptoss20b-en-en	0.4	0.528	0.235	0.234	0.338	0.508	0	0.362
gemini35-en-fr	0.4	0.694	0.255	0.242	0.263	0.485	0.167	0.349
qwen332b-en-fr	0.222	0.512	0.163	0.165	0.286	0.615	0	0.334
qwen3235b-fr-fr	0.19	0.407	0.32	0.243	0.319	0.369	0.087	0.33
gemini35-en-en	0.364	0.652	0.222	0.233	0.23	0.5	0	0.324
qwen332b-fr-fr	0.333	0.558	0.125	0.177	0.236	0.632	0	0.315
qwen3235b-en-fr	0.182	0.321	0.24	0.254	0.35	0.339	0.091	0.315
qwen3235b-en-en	0.182	0.275	0.302	0.227	0.279	0.408	0.133	0.299
llama318b-en-fr	0.111	0.12	0.192	0.281	0.319	0.507	0.2	0.297
lsd-fr	0	0	0.253	0.289	0.479	0.057	0	0.29
llama4scout-fr-fr	0.125	0.167	0.182	0.291	0.341	0.356	0.1	0.286
lsd-en	0	0	0.267	0.231	0.472	0	0	0.276
llama318b-fr-fr	0.143	0.197	0.179	0.174	0.239	0.543	0	0.267
llama323b-fr-fr	0	0.364	0.2	0.195	0.24	0.395	0	0.266
qwen332b-en-en	0.222	0.364	0.172	0.19	0.168	0.567	0.333	0.261
llama318b-en-en	0.138	0.222	0.136	0.265	0.246	0.479	0.143	0.261
llama4scout-en-fr	0.154	0.261	0.143	0.258	0.281	0.329	0.08	0.257
llama4scout-en-en	0.105	0.28	0.125	0.246	0.234	0.434	0.154	0.252